Inside the Box, Then the Contract

ME/CE/AM 295

Agentic AI in Scientific Discovery

9 units (3-5-1)

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Lecture 3

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The Engine Under the Hood



From Language Models to Autonomous Agents

WHERE WE LEFT OFF · MONDAY

The model never sees a number, and “close” means “co-occurs”. Today: what happens between the tokens going in and the token coming out — and how to make that output a contract.

PART 1

The Engine

—

PART 2

The Contract

tokens in → attention → one token out → declared shape

One Token at a Time

Similar to time-stepping in FEA to read a full sentence

Every pass reads the whole context and emits one token. Append it, and the pass repeats.



Where the instruction goes
attention is most reliable at the start of the context

Why the window is finite
every token scores every other token — n^2 work

once infinite, AGI becomes true

Why the middle gets lost
the model reads everything, but not evenly

Why a scratchpad helps
a pass is the unit of thought — Monday

One fact, four consequences. The four cards are the map for today and Monday.

NB Memory of AI is not allocated until you tell it to remember that

The Transformer (2017)

Vaswani et al., “Attention Is All You Need”. One architectural decision made the scaling of the last decade possible.

BEFORE 2017 — RNNs, LSTMs

Read text one token at a time, carrying a hidden state forward

Slow to train, hard to parallelise, and a token at the end has only a faded memory of the beginning.

2017 — THE TRANSFORMER

Every token attends to every other token at once, whatever the distance.

All positions computed in parallel → trains on GPU clusters over the whole internet.

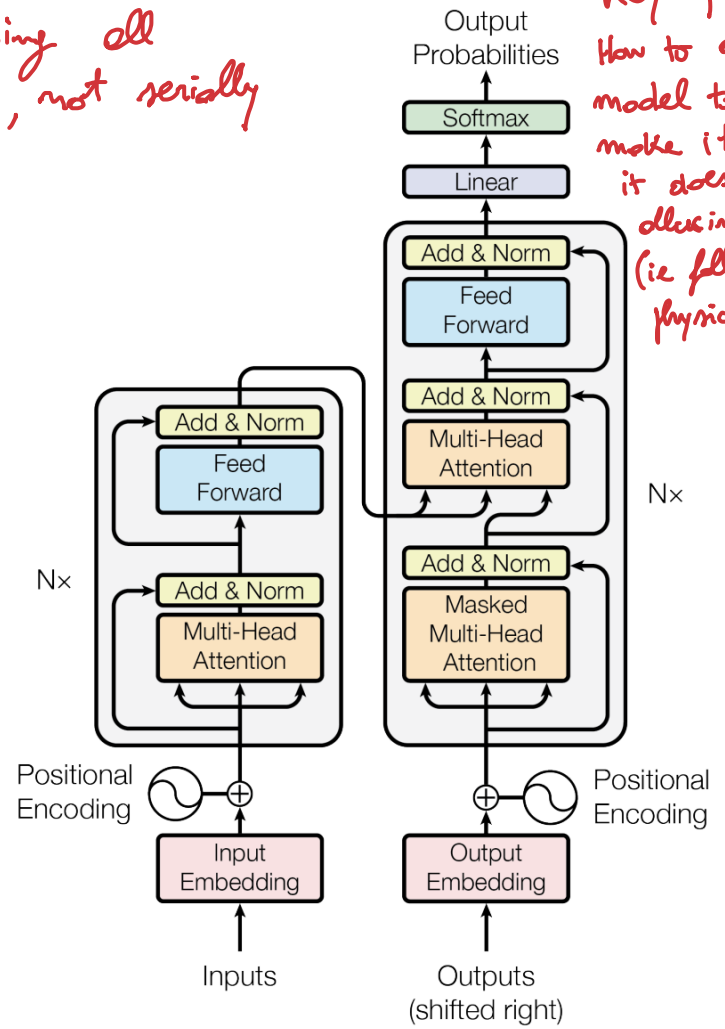


Figure 1: The Transformer - model architecture.

Why you should care: the thing that made the model trainable at scale is the same thing that limits its context and shapes what it attends to.

Vaswani et al., NeurIPS 30 (2017), arXiv:1706.03762

Self-Attention — One Head

relationship between 2 words, no matter where they are

For every token, three projections of its embedding — and one table of weights per head.

QUERY Q

What this token is looking for in its context.

the search string you type

KEY K

What this token advertises about itself. (information around token)

the index entries in the catalogue

VALUE V

The content this token actually carries.

the record you get back on a match

THE CORE MECHANISM

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

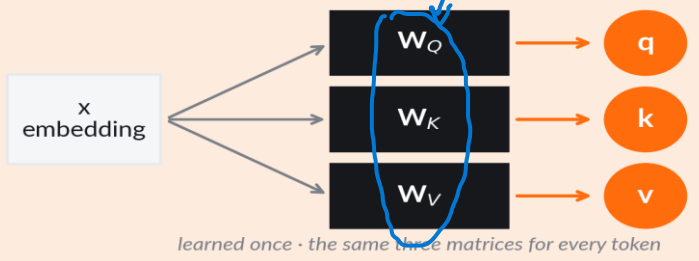
$Q \cdot K^T$ scores every pair of tokens · softmax turns scores into weights that sum to one · the result is a weighted blend of Values

score = $Q \cdot K^T$

normalization factor

WHERE Q, K, V COME FROM

rectangular matrices



($1/28$ in all models)

Self-check: which three vectors are computed for every token? → Query, Key, Value

For each iteration, we calculate all three projection matrices

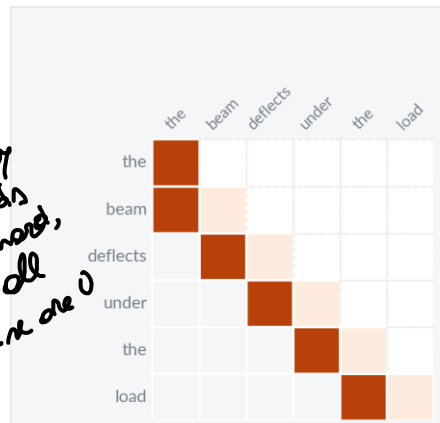
Many Heads, One Layer

Each head has its own W_Q, W_K, W_V — the same sentence gets many tables of weights at once.



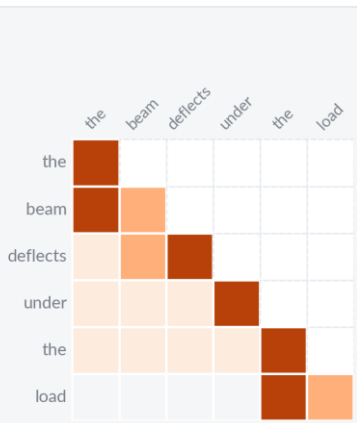
head 12: verb → subject

"deflects" reads "beam"



head 40: previous token

every token reads the one before



head 71: noun → article

"beam", "load" read "the"

Same input, same layer — three of ~96 tables. Heads specialise; their outputs are concatenated and passed on.

THE BUDGET

96 heads × 128 = 12 288 = the width.
Same parameter count as one wide head, spent as 96 narrow views. 128 = d_k, the number under the square root.

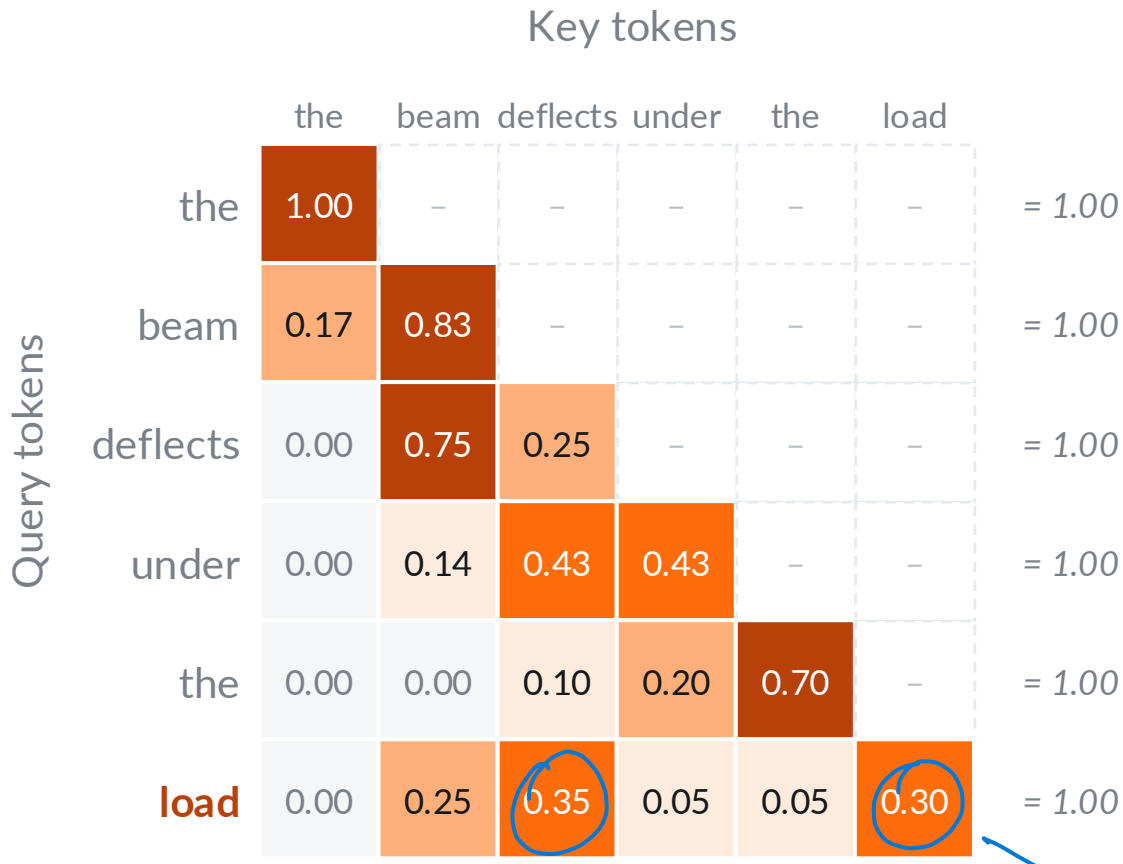
	width d	heads	d_k
Transformer '17	512	8	64
GPT-3	12 288	96	128
Llama 3 70B	8 192	64	128

Chosen, not derived: more heads at larger width, each kept 64–128 wide.

Analogy: a strain-gauge rosette — one gauge gives one direction; several orientations reconstruct the full state at a point.

Attention in Action

One head, one sentence: “the beam deflects under the load”



– *masked: a token cannot attend to what comes after it (causal attention)*

high attention given to load - deflects , load - load

READ THE “LOAD” ROW

By the last token the model reads back to “deflects” and “beam” — the verb and its subject — and blends their Values into “load”. “Deflects”, two tokens earlier, could only see “the beam”.

WHY DIVIDE BY $\sqrt{d_k}$

Dot products grow with dimension. Without the scaling the softmax saturates onto one key and gradients vanish. Scaling is non-dimensionalisation — the same reason you scale before you solve.

Illustrative weights, not from a real model. Causal mask as in decoder-only models (Claude, GPT); encoders like BERT fill the upper triangle.

Reading vs. Inference — One Machine, Two Ways to Run It

The causal mask lets training process a whole sequence at once. Generation still goes one token at a time.

READING — ONE PARALLEL PASS

each position predicts the token to its right — all at once

the	beam	deflects	under	the	load

Feed the whole document. Every position predicts the token after it, simultaneously; the mask hides the future. One pass, one loss, one gradient step — on thousands of GPUs.

INFERENCE — ONE TOKEN PER PASS

pass 6 waits for pass 5

the	beam	deflects	under	the	?

The next token depends on the one just chosen, so generation cannot be parallelised. Read in parallel, write serially.

n	predictions per training pass
1	token per inference pass
$	input tokens cheap, output tokens dear — Monday's cost model

Why it matters: parallel training is the mechanism behind Chapter 1's scaling argument — it is what let this architecture absorb compute. Slide 5 was about reading; slide 4 was about writing. Both are true.

Vaswani et al. 2017, §3.2.3 (masking) · Chapter 1 §1.4.3 (the scaling argument)

Positional Encoding — Attention Alone Has No Word Order

Self-attention is permutation-invariant. It sees a bag of tokens.

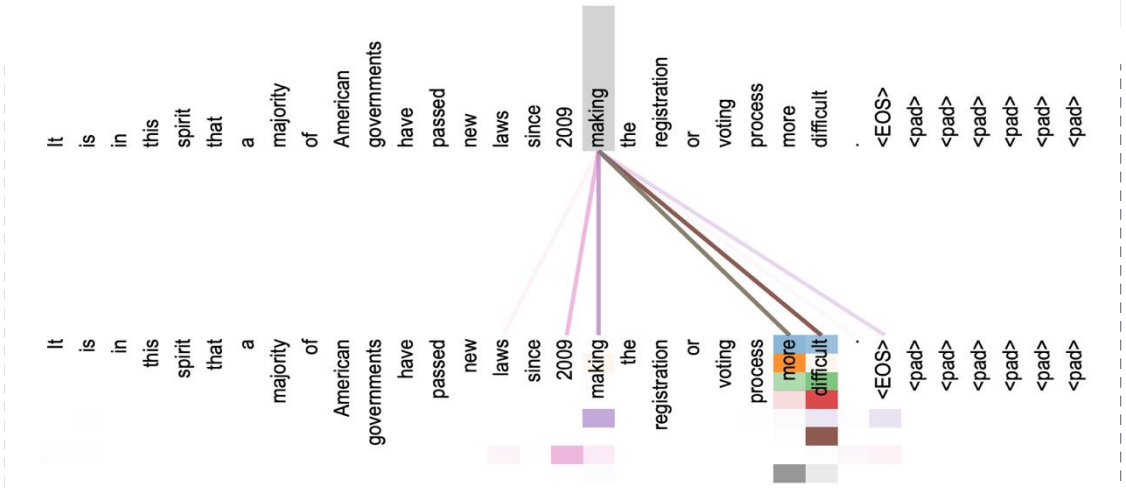
"the beam supports the load"
"the load supports the beam"

identical to attention without position

- The fix: add a position vector to every token embedding before attention runs.
- Sinusoidal in the original paper; learned after; rotary (RoPE) in most current models.

ENGINEERING ANALOGY — NODE COORDINATES

A connectivity table tells you which nodes touch. Without node coordinates it is not a mesh — it is a graph. Position vectors are the coordinates of the sequence.



Vaswani et al., NeurIPS 30 (2017), arXiv:1706.03762

Context Windows — More Tokens, Quadratically More Work

Every token scores every other token. Double the prompt → four times the attention compute.

~2 K

tokens · 2020

GPT-3: a few pages

128 K

tokens · 2023

a 300-page book

1–2 M

tokens · 2025–26

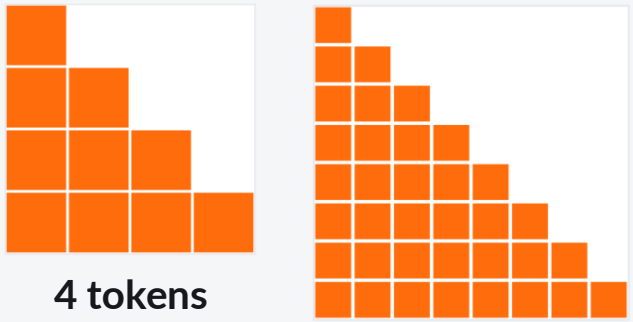
a shelf of standards

THE n^2 WALL

Score matrix: n^2 numbers to compute and, done naively, n^2 numbers to store and move through GPU memory. At 100 K tokens that is 10^{10} scores per head, per layer, per pass.
Two escapes — next slide.

for too much computation

WHY n^2



4 tokens
10 scores

8 tokens
36 scores

Every token scores every earlier one. Double n, roughly quadruple the scores — and the memory to hold them.

Window sizes are representative published values; the frontier entry is indicative of 2025–26 models.

Three Ways to Pay for Attention

Dense and sliding window decide what the model computes. FlashAttention decides how.

DENSE



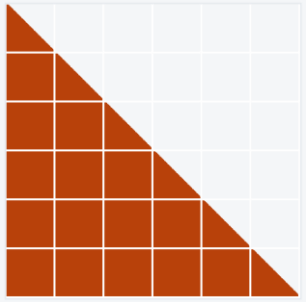
every token scores every earlier token

n^2 compute · n^2 memory

exact — the reference

≈ the assembled stiffness matrix

FLASHATTENTION



same scores, computed in tiles that stay on-chip; the n×n matrix is never written out

n^2 compute · n memory

discretize the attention matrix

exact — same answer, faster

≈ a matrix-free solver

SLIDING WINDOW



each token sees only its last w neighbours; far scores are not computed at all

n·w compute · n·w memory

an approximation — a different model, trained that way

≈ the banded stiffness matrix

FlashAttention: a better way to compute the same thing. Sliding window: a cheaper thing to compute. Frontier models use the Flash-class kernel for both kinds of layer.

Who Uses What — and Why It Matters to You

Where the layout is public, modern models mix dense and windowed layers. The ones you call do not say.

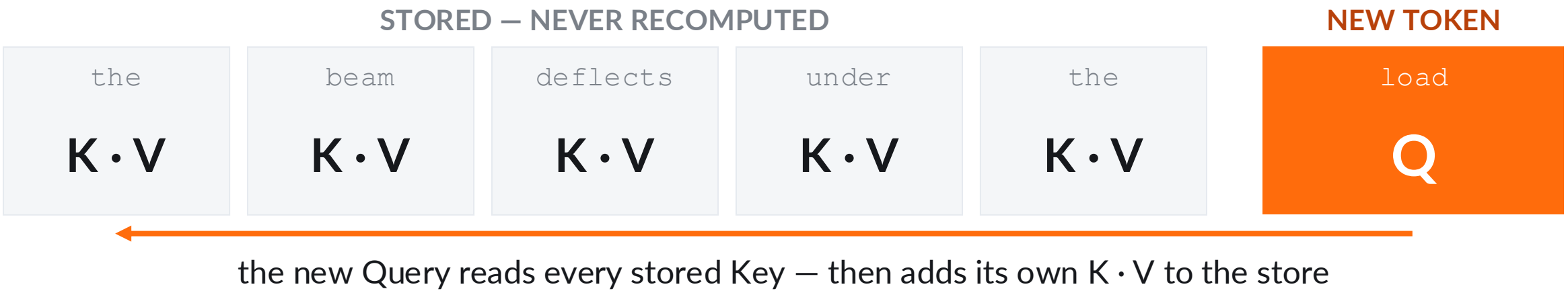
Model	Year	Attention layout	Context
GPT-3	2020	alternating dense and banded-sparse layers	2 K
Llama 2 / 3	2023–24	dense in every layer	4 K → 128 K
Mistral 7B	2023	sliding window (4 K) in every layer	32 K
Gemma 2 / 3	2024–25	1 local : 1 global → 5 local : 1 global	8 K → 128 K
gpt-oss	2025	alternating windowed (128) and dense layers	128 K
Claude · GPT · Gemini	current	not published	200 K – 2 M

The point: a two-million-token window is a storage figure, not a promise that every token is read evenly — the next slide measures it.

Brown et al. 2020 · Llama 2/3 papers · Jiang et al. 2023 · Gemma 2 & 3 reports · gpt-oss model card 2025 · frontier layouts unpublished; verify before class

The KV Cache — Why a Pass Is Not a Full Recompute

Keys and Values are computed once and kept. Only the new Query reads the whole context.



cache memory

∝ n

MB per token; 100 K tokens is hundreds of GB

compute per token

∝ n

one new Query against n Keys; long chats get slower and dearer

prompt caching

keep the prefix's K·V between calls; pay less for unchanged input

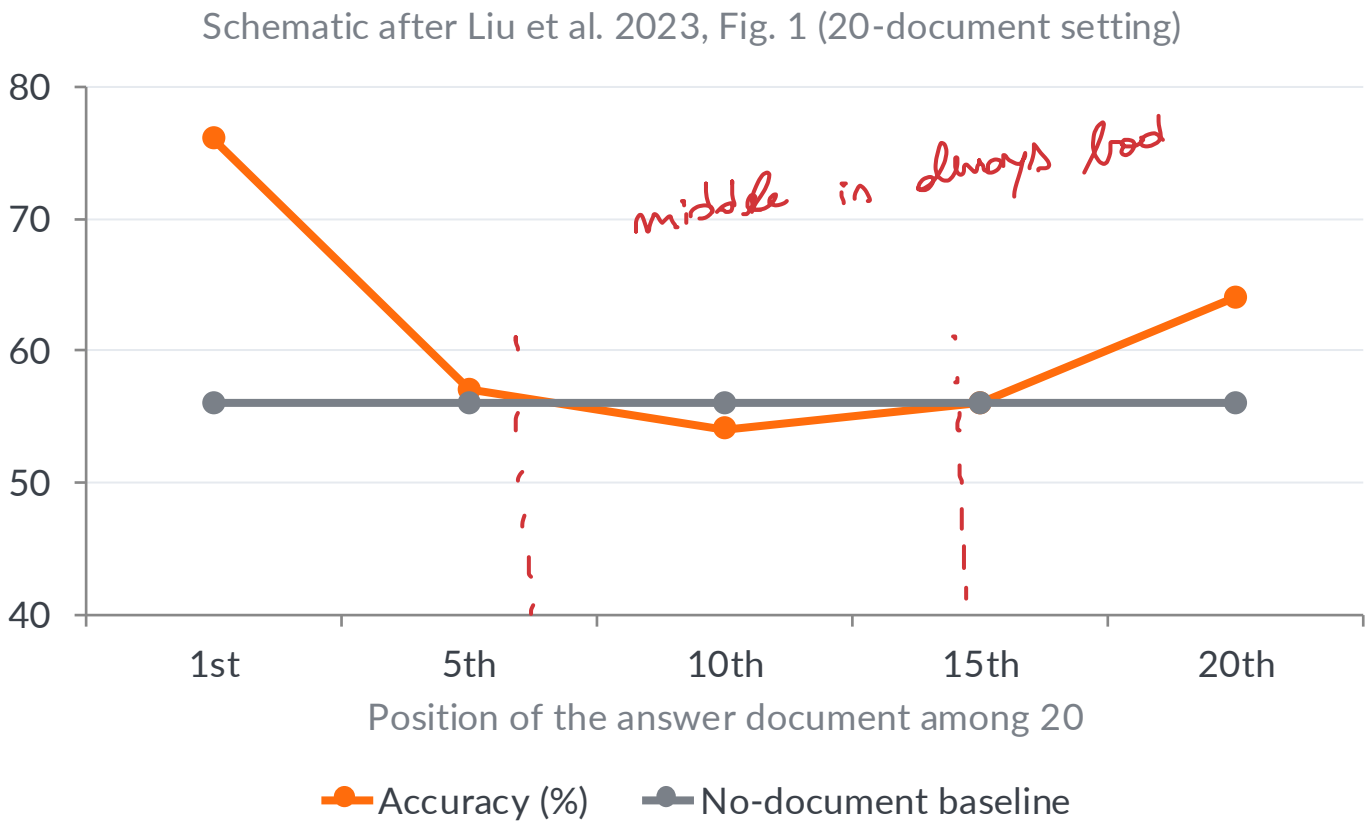
Analogy: adding a node to an assembled mesh — keep the stored matrix, compute one new row and column.

Pope et al. (2022), arXiv:2211.05102 · Anthropic, Prompt caching guide · GPT-3 dimensions: Brown et al. 2020

NB Must constrain the compression, aka tell the AI what is important to remember

Bigger Context ≠ Better Retrieval

Liu et al. 2023: move the one document that holds the answer through 20 retrieved documents.



THE U-SHAPE

Highest when the answer is first or last. In the middle it can fall below the model's accuracy with no documents at all.

ENGINEERING IMPLICATION

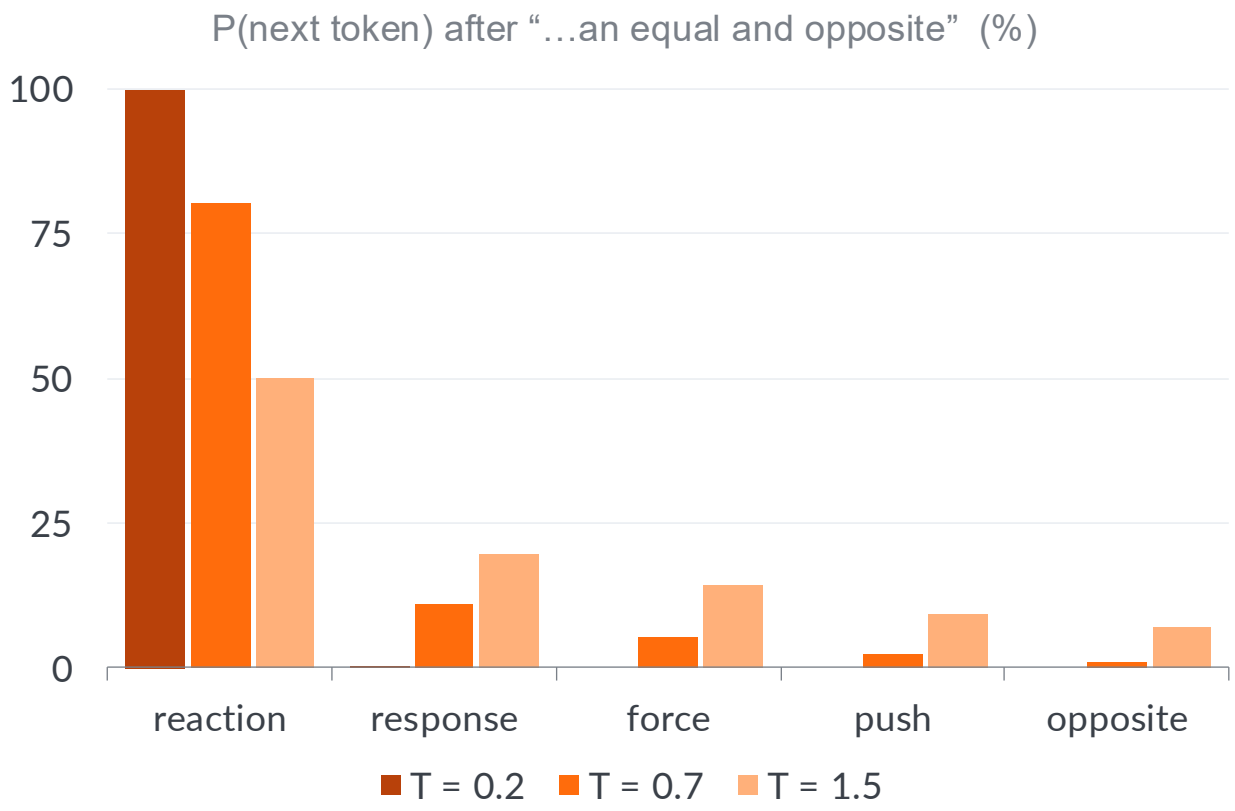
Pasting an ASM Handbook into the prompt is legal and unreliable: a coefficient on page 400 may be silently missed. Week 4 (RAG) retrieves the relevant chunks and puts them first.

Liu et al., "Lost in the Middle", TACL 12 (2024), arXiv:2307.03172 · Recommended reading

This gets fixed by shortening the context so it can understand the whole thing

Sampling — the Softmax You Control

The last layer gives a distribution over the next token. Temperature rescales it before you draw.



TWO SOFTMAXES, ONE EQUATION

	Attention	Sampling
lives	in the model, each head	after the model, per pass
acts on	$Q \cdot K^T$ scores	next-token logits
temp.	$\sqrt{d_k}$ — fixed at training	T — you set it per call
saturated	one key wins — stuck	one token wins — T = 0

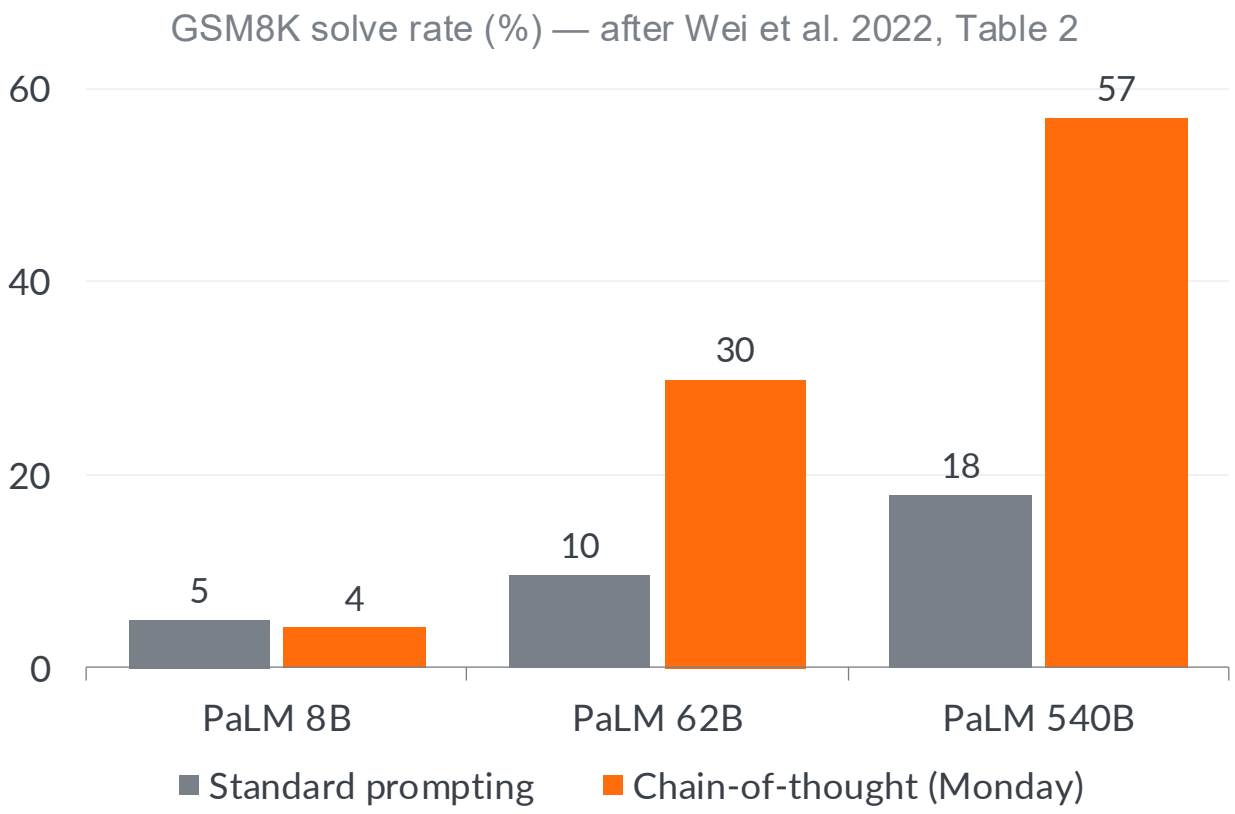
ENGINEERING RULE OF THUMB

T = 0 for extraction and arithmetic — lowest variance, not a determinism guarantee. **T ≈ 0.7** for design ideation. **T > 0** whenever you will vote over several runs — Monday.

Top-p / top-k prune the tail before T acts. Logits illustrative. Note: current models of the course provider accept only the default T — the concept is what matters; reproducibility here comes from a typed contract + a pinned model ID.

Scale Unlocks Abilities — or Does It?

Abilities absent in small models that appear suddenly past a size threshold — or a metric artefact.



ANALOGY — A GLASS TRANSITION

Nothing in the formula changes. Cross a threshold and the behaviour is qualitatively different.

THE CAVEAT — SCHAEFFER ET AL. 2023

Much of the sharpness is the metric: all-or-nothing scoring turns a smooth gain into a cliff. Measure partial credit and the curve bends gently.

WHAT IT MEANS FOR YOU

Test the model you will deploy, at the size you will deploy it. A small local model can fail where a frontier model succeeds — same architecture, different regime.

Wei, Tay et al., TMLR 2022 (arXiv:2206.07682) · Schaeffer, Miranda & Koyejo, NeurIPS 2023 (arXiv:2304.15004) · values from Wei et al. 2022 (arXiv:2201.11903), Table 2

What the Engine Tells You

Five facts about the mechanism, five rules for using it. Four today, one on Monday.

1 **Put the instruction where attention is reliable**

Instructions and format spec first, data in the middle, question last.

2 **Treat the context as a budget, not a bucket**

n^2 work, and the middle degrades. Send what matters, not everything you have.

3 **Position is injected; order is yours to set**

The model knows order only because you put it there. Keep documents whole and in reading order.

4 **Test the model you will deploy, at the size you will deploy it**

Abilities appear with scale and T is yours to set: benchmark the exact model and settings you ship; log the model ID.

5 **A pass is the unit of thought**

One token out per full read. A model forced to answer at once has thought once.

Part 1 ends here. Monday finished with “a prompt is a boundary condition”; now you know where the boundary is most rigid.

BRIDGE · FROM THE ENGINE TO THE CONTRACT

The model reads everything, attends unevenly, and emits one token at a time.
So: put the instructions where attention is reliable — and declare the shape of the answer instead of hoping for it.

PART 1

The Engine

—

PART 2

The Contract

roles → three ways to pin a shape → schema → validation

Where Your Words Land — Typed by You or by an Agent

Three roles, three places in the context. In your own agents you set them; in Claude Code you should know why.

`system`

Your standing instructions: persona, rules, units, the null rule. Set once per pipeline — and placed at the start of the context, where attention is most reliable.

`user`

The task and the data for this call: the abstract, the beam parameters, the question. Untrusted content goes here — never in the system prompt.

`assistant`

The model's previous turns. Monday's few-shot examples were these; in agents they carry tool calls.

```
from anthropic import Anthropic
client = Anthropic()

resp = client.messages.parse(
  model=MODEL, max_tokens=1024,
  system=(   # standing instructions
    "You extract material properties.
     Units: MPa, GPa, kg/m^3.
     If a value is not reported,
     return null. Never guess."),
  messages=[{"role": "user",   # data
             "content": abstract_text}],
  output_format=MaterialProperties)
```

← instructions before data: Monday's rule, today's mechanism

Schema + system prompt together: the schema forces the shape; the system prompt says what goes in it and what to leave null.

Three Ways to Pin a Shape

Monday ended on “when the shape must always hold, declare it”. The ladder, and what each rung costs.

1	**DESCRIBE IT** Rule 3: “Return JSON with fields material, grade, yield_mpa.”	Costs nothing. Fails quietly: three runs, three shapes — 634 MPa, “approximately 634 megapascals”, YS = 634 ± 9.	*Week 1*
2	**SHOW IT** Few-shot: examples in the context raise the odds of the pattern — they do not enforce it.	~360 tokens on every call, forever. Pins only the shapes you showed: two alloys, N phases and the empty case all have to be in the examples.	*Monday*
3	**DECLARE IT** A schema compiled into a grammar: tokens outside it get probability zero. Valid every time.	Costs one class definition. Fails where no shape can help: the numbers inside it can still be physically wrong.	*Today*

```
props = resp.parsed_output
props.yield_strength_mpa   # 250.0, a float
```

A solver does not negotiate. It needs rung 3 — and rung 3 is where today’s real work begins.

Rung 1 is Chapter 1 §1.9.2; rung 2 is Monday’s zero-/few-shot slides; rung 3 is the rest of today.

In prompt, you must pass an unbreakable rule that it must consider and do before each loop iteration is performed. This extra layer is expensive but suppresses incorrect output

Declare the Shape — the Contract Every Tool Will Use

The agent will write this class. Here is how to tell whether it wrote it well.

```
from typing import Optional
from pydantic import BaseModel, Field

class MaterialProperties(BaseModel):
    material_name: str = Field(
        description="Name of the material")
    yield_strength_mpa: Optional[float] = Field(
        None, description="Yield strength in MPa;
        null if not reported")
    youngs_modulus_gpa: Optional[float] = Field(
        None, description="Young's modulus in GPa")
    is_isotropic: Optional[bool] = None

resp = client.messages.parse(model=MODEL,
    max_tokens=1024, system=SYSTEM,
    messages=[{"role": "user", "content": abstract}],
    output_format=MaterialProperties)
props = resp.parsed_output   # typed object
```

Units live in the field name

yield_strength_mpa, not yield_strength. Three places the model reads the unit: field name, description, system prompt.

Measured fields are Optional

A required field on an unreported quantity is a request for a hallucination.

Descriptions are instructions

The model reads them — like the tool docstrings you meet next week.

Anthropic, Structured outputs guide — docs.claude.com (required reading) · Pydantic docs: docs.pydantic.dev

Schema-Valid ≠ Physically Valid

Pydantic does not fix hallucinations. It types them.

✓ PASSES THE SCHEMA — every field the right type

```
props = resp.parsed_output

props.material_name        # "AISI 4140"
props.yield_strength_mpa   # 25000.0   float
props.youngs_modulus_gpa   # 0.21      float
props.density_kg_m3        # 7850.0    float
```

✗ FAILS PHYSICS

25 000 MPa yield
the paper said 25 ksi (172 MPa); the model converted to psi, not MPa.

0.21 GPa modulus
210 GPa mistranscribed into the wrong unit field.

Both are floats. The parse succeeded. Pydantic's type check is happy. Your simulation runs — and it is wrong.

Types check shape, not physics. The next slide adds the check that does.

This is the slide that separates people who have shipped an extraction pipeline from people who have read about one.

Validators — an ASTM Spec for Data

Right type, right range, internally consistent — or rejected at the gate.

```python
from pydantic import field_validator, model_validator

class MaterialProperties(BaseModel):
    ...
    @field_validator("yield_strength_mpa")
    @classmethod
    def plausible(cls, v):
        if v is not None and not (100 <= v <= 2500):
            raise ValueError("yield out of range: MPa?")
        return v

    @model_validator(mode="after")
    def consistent(self):
        ys, uts = self.yield_strength_mpa, self.uts_mpa
        if ys and uts and ys > uts:
            raise ValueError("yield > UTS")
        return self
```

Per-field bounds

steel yield 100–2 500 MPa · E 1–1 200 GPa · ρ 500–25 000 kg/m³. A unit slip fails loudly.

Cross-field physics

yield < UTS · E/ρ within the material class. A model_validator catches inconsistent pairs.

Retry, then log — yours to write

The agent cannot know 2 000 MPa is real for maraging steel. Bounds are your judgment. Re-ask once; still failing → log, don't drop.

A validated schema is an ASTM spec for data: the material is accepted or rejected at the gate, with the reason written down.

Where the JSON Goes — Revisited

Monday's pipeline, with stage 2 now filled in. Extraction is still stage 3 of 5.



INSIDE STAGE 2 — WHAT YOU MUST BE ABLE TO READ

`system prompt`	→ `user message`	→ `schema`	→ `validators`	→ `retry` → `log`
units · null rule · never guess	the abstract, untrusted	fields, types, Optional	bounds · cross-field	one re-ask, then a record

A solver does not negotiate. Now a null stays a null, a unit slip is rejected, and the record says why.

Lecture 3 — Takeaways, Readings, and Monday

The engine

One token per pass; every pass reads the whole context. Attention scores every pair (n^2), position is injected, the middle of a long prompt is read least reliably.

The contract

Roles put instructions where attention is reliable. A Pydantic schema declares the shape and the API enforces it. Validators turn the schema into a spec that checks physics.

The thesis, again

The instrument that generates a result cannot be the only thing that validates it — today at the scale of one JSON field.

The open thread for Monday: the schema declares a shape, but the model can still be wrong inside it — it cannot multiply. Bring that question.

READINGS (COURSE SITE)

Required Vaswani et al. 2017, §3 only · Anthropic, Structured outputs guide

Recommended Liu et al. 2023, Lost in the Middle · Pydantic validators docs

Optional Wei, Tay et al. 2022 + Schaeffer et al. 2023, emergent abilities and the "mirage" rebuttal — read as a pair

MONDAY · LECTURE 4

Reasoning you can audit, at a price you can estimate: chain-of-thought, self-consistency, tree-of-thoughts, the cost model — and the reasoning field that goes first in your schema.

Discussion points (optional, not assessed): Q-K-V · the n^2 bottleneck · lost in the middle · why measured fields are Optional · schema-valid ≠ physically valid

References — Lecture 3

Full citations for the works cited in this deck (all on the course site under References/)

Ref_41 Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., Polosukhin, I. (2017). Attention Is All You Need. NeurIPS 30. arXiv:1706.03762

Ref_47 Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., Liang, P. (2024). Lost in the Middle: How Language Models Use Long Contexts. TACL 12, 157–173. arXiv:2307.03172

Ref_51 Beeravolu, A. R., Kishore Babu, N., Talari, M. K., Rehman, A. U., Srirangam, P. (2023). Influence of Microstructure and Mechanical Properties of Dissimilar Rotary Friction Welded Inconel to Stainless Steel Joints. Materials 16(8), 3049. CC BY 4.0

Ref_52 Wei, J., Tay, Y., Bommasani, R., et al. (2022). Emergent Abilities of Large Language Models. TMLR. arXiv:2206.07682

Ref_53 Schaeffer, R., Miranda, B., Koyejo, S. (2023). Are Emergent Abilities of Large Language Models a Mirage? NeurIPS 36. arXiv:2304.15004

Ref_54 Dao, T., Fu, D. Y., Ermon, S., Rudra, A., Ré, C. (2022). FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. NeurIPS 35. arXiv:2205.14135

Ref_55 Anthropic. Structured outputs — developer guide. platform.claude.com/docs/en/build-with-claude/structured-outputs (checked Oct 2026)

— Pydantic documentation — validators. docs.pydantic.dev/latest/concepts/validators/ · GSM8K values on the emergent-abilities slide: Wei et al. (2022), arXiv:2201.11903, Table 2

— Pope, R., Douglas, S., Chowdhery, A., et al. (2022). Efficiently Scaling Transformer Inference. arXiv:2211.05102 (the KV cache and its memory cost) · Anthropic, Prompt caching — developer guide, platform.claude.com